Type 1: Research Project | Phase 1 Review 2

Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation

Team Number: 28

S.No	Register Number	Student Name
1	CB.SC.U4CSE23363	Haswitha K
2	CB.SC.U4CSE23529	Hasini Reddy M
3	CB.SC.U4CSE23547	Sheela Akshar Sakhi
4	CB.SC.U4CSE23761	Kousik Sarma Lakkaraju
5	CB.SC.U4CSE23753	V.Chakravarthy

Guide: **Dr. G R RAMYA, Dr. VANDHANA S**

Department of CSE

Project Proposal and Objectives

Overall Project Idea: Instead of implementing standard off-the-shelf ML algorithms, our team is proposing a **novel mathematical extension** to existing FL frameworks called the **Federated Personalized Drift-Aware Attention Framework (FPDAF)**. This custom algorithm modifies standard *FedAvg* and *CUSUM* to directly target Multivariate Healthcare Time-Series.

Objectives to Achieve

1. **Privacy Preservation:** Train predictive models across multiple hospitals collaboratively without sharing raw patient data (Federated Learning).
2. **Personalization:** Use Local Personalization Heads to manage patient heterogeneity (Non-IID distributions).
3. **Drift Adaptation:** Implement CUSUM algorithms to detect when a patient's physiological state degrades and rapidly adapt the model locally.
4. **Clinical Trust:** Provide transparency using Temporal Attention mechanisms so doctors know *why* the model predicted a risk.

Addressing Review 1 Panel Comments

Review 1 Panel Feedback	**Adjustments Made for Review 2**
1. Define application scope strictly	Narrowed domain completely out of generalized data into **Healthcare ICU Time-Series (Sepsis / Deterioration Profiling)**.
2. Target a well-defined dataset	Selected the **PhysioNet Challenge 2019 Dataset** (40,000+ real multivariate ICU records).
3. Articulate research novelty & Drift	Defined the exact **CUSUM mathematical mechanism** triggering dynamic retraining when a patient's physiological state aggressively changes.
4. Improve technical depth & XAI	Designed a formal split-architecture using **Temporal Attention** so clinicians visually see *which* vital signs cause risk alerts.
5. Comparative FL Discussion	Contrasted generalized *FedAvg* with our localized *FedPer* nodes to prove handling of hospital heterogeneity.

Introduction and Background

Introduction to the Problem: In modern Intensive Care Units (ICUs), continuous monitoring systems generate vast amounts of high-frequency multivariate time-series data (Heart Rate, Blood Pressure, SpO2). Detecting critical deterioration early saves lives.

Motivation to Solve This Specific Problem:

- **The Data Silo Problem:** High-accuracy Deep Learning models require massive datasets, but rigid healthcare privacy laws (e.g., HIPAA) forbid combining data into a centralized server.
- **The Global Model Failure:** If multiple hospitals successfully engage in standard Federated Learning (FedAvg), the resulting "global model" fails because hospital demographics and local treatment protocols vary wildly.
- **The Black Box Problem:** Doctors cannot act on an AI alert if the model cannot explain the physiological basis for its warning.

Research Gap Analysis

Existing Research	Identified Limitations / Gap	Need / Why Gap Matters	Project Contribution
Centralized ML and on-device analytics systems (Papers 1, 17, 19, 23)	Do not support federated learning or privacy-preserving collaboration	Centralized data sharing violates privacy laws and is unsuitable for distributed environments	Enables collaborative analytics using federated learning without sharing raw data
Basic Federated Learning frameworks (FedAvg) (Papers 2, 4, 5)	Lack personalization, explainability, and analytics-level insights	A single global model performs poorly on heterogeneous client data	Introduces personalized local models with global aggregation and analytics
Personalized Federated Learning methods (FedPer, pFL) (Papers 3, 10, 16, 20)	Personalization exists but lacks explainability and drift awareness	Without explainability and adaptation, predictions cannot be trusted over time	Combines personalization with explainable analytics and drift handling
Concept driftaware FL approaches (Papers 2, 9)	Drift handling is not integrated with analytics or forecasting pipelines	Real-world data behavior changes over time, degrading prediction accuracy	Integrates drift detection with time-series analytics and forecasting
Federated Time-Series Forecasting works (Papers 6, 8, 1015, 18)	Primarily focus on accuracy; limited explainability and personalization	Stakeholders require transparency to trust and act on predictions	Provides explainable insights alongside federated forecasting
Explainable AI approaches (Papers 7, 17, 19, 23)	Explainability mostly exists in centralized or non-time-series settings	Explainability is critical for trust in sensitive domains like healthcare and energy	Adapts explainable AI techniques to federated time-series analytics
Healthcare-focused FL applications (Papers 9, 21, 22, 24, 25)	Application-specific solutions with limited generalization	A reusable framework is required across multiple domains	Proposes a domain-agnostic federated analytics framework
Security- and privacy-enhanced FL methods (Papers 13, 24, 25)	High complexity and low usability despite strong privacy guarantees	Practical systems must balance privacy, performance, and usability	Balances privacy, efficiency, and interpretability in a single framework

Dataset: PhysioNet Challenge 2019 (Sepsis Prediction)

Dataset Overview:

- **Source:** PhysioNet (Public Healthcare Dataset).
- **Data Type:** Real-world multivariate ICU patient monitoring time-series.
- **Structure:** 40,000 patient files (`.psv`), with each row representing one hourly observation.
- **Key Features:** Heart Rate, BP, SpO2, Temperature, ICULOS.
- **Target:** `SepsisLabel` (predicting deterioration).

Why PhysioNet?

- **Heterogeneity:** `setA` & `setB` come from different hospitals (perfect for FL).
- **Time-Series:** Sequences support LSTM & Drift tracking.
- **Clinical:** Interpretable vitals support our XAI mechanism.

Data Preprocessing & Imputation Pipeline

Challenge & Why it looks like a mess: Real clinical records in PhysioNet contain massive sparsity and `NaN` gaps. High-frequency vitals (Heart Rate) are logged continuously by bedside monitors, while low-frequency lab tests (Bilirubin, Calcium) are only drawn manually by nurses every 12-24 hours.

Engineered PyTorch Pipeline: We developed a local Python preprocessing engine to dynamically ingest the 40,000+ '.psv' files and render clean PyTorch Tensors:

1. **Forward-Filling (Time-Series Carry):** Carries the last-known medical lab value forward in the sequence until a new blood measurement is taken.
2. **Backward-Filling & Zero-Padding:** Synthesizes initial missing conditions and replaces completely unmeasured lab variables with 0s to prevent Neural Network failure.
3. **Tensor Standardization:** Aggregates parameters via `StandardScaler` so highly varying units (e.g., Blood Pressure vs pH levels) do not skew the FPDAF gradient descent.

Fundamentals Learned in the Semester

To execute the plan above, our team embarked on extensive upskilling over this semester:

- **Federated Learning Architecture:** Extensively reviewed *McMahan et al.'s* fundamental papers to understand gradient averaging and the shift from Centralized ML to Decentralized ML.
- **PyTorch Framework Fundamentals:** Completed MOOCs & online tutorials to transition our team's technical skills into PyTorch (specifically custom `nn.Module` networks, `DataLoaders`, and customized training loops).
- **Time-Series Analytics:** Studied LSTM sequence modeling and Attention-mechanism math (computing context vectors via softmax weights).
- **Statistical Process Control:** Learned the *CUSUM* (Cumulative Sum) statistical methodology for detecting small shifts in ongoing time-series data.

Mathematical Foundation

1. Standard Federated Averaging (FedAvg Base):

$$w_{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} w_{t+1}^k \tag{1}$$

- Server aggregates weights w_{t+1}^k from K hospitals weighted by patient count n_k. (Personalization heads are strictly excluded from this sum).

2. CUSUM Drift Detection (Client-Side Trigger):

$$S_t = \max(0, S_{t-1} + (e_t - \mu - k)) \tag{2}$$

- e_t: Prediction error (residual) at time step t.
- μ: Mean error during stable conditions.
- If the cumulative statistic $S_t > H$ (Threshold), a concept drift is flagged, triggering local model retraining.

Algorithmic Novelty & Innovation Plan

The Research Novelty of FPDAF: We are not utilizing standard implementations of *FedAvg* or *CUSUM*. We are proposing a mathematical extension that fuses them into a single, responsive clinical algorithm.

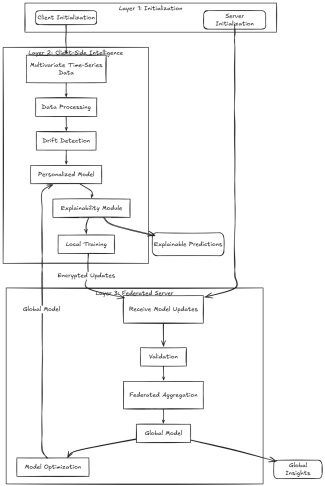
The Novel Plan:

- **CUSUM as a Neural Trigger:** We mathematically repurpose *CUSUM* from a simple statistical quality-control tool into an active Neural Network manipulator. When the formulation $S_t > H$ alerts, our novel code aggressively unfreezes the isolated Local Personalization Head, forcing immediate recalibration without destroying the global backbone.
- **Attention-Weighted Decentralization:** Standard *FedAvg* treats all hospitals equally based on sample size. FPDAF will introduce a novel extension aggregating weights locally, prioritized by our *Attention Mechanism*'s confidence, preventing outlier clinical anomalies from polarizing the central server.

Architecture Diagram and Explanation of Design

Why this design?

- **Privacy:** Only encrypted gradients travel upward.
- **Local LSTM Base:** Captures generalized physiological rules.
- **Attention Layer:** Provides XAI (feature importance) right before the local prediction.
- **Personalization Head:** NEVER shared with the server. Tailors risk predictions to the specific local patient cohort.



Summary of Works Completed (Technical Evidence)

Phase 1 Contributions mapping to Technical Validation & Evidence:

1. **Problem Formulation:** Successfully scoped the application domain specifically to healthcare ICU telemetry.
2. **Mathematical Formulation:** Defined formulas for exactly how CUSUM and Attention will be mathematically integrated into the *FedPer* pipeline.
3. **Architecture Design:** Finalized the multi-layered system diagram.
4. **Repository Framework Setup (Baseline Code):**
   - Initialized project GitHub repository (`https://github.com/aksharsakhi/23CSE399-Project-Phase-1`).
   - Pulled and set up the foundational **FedAvg-PyTorch base framework**.
   - Built initial custom PyTorch stubs simulating the isolated ‘LocalPersonalizedModel‘ and a synthetic ‘SimulatedICUDataset‘ proxy.

Plans for the Oncoming Semesters (Phase 2)

The direct roadmap to full implementation:

Target Area	**Objective for Phase 2**
Data Preprocessing	Implement normalization, handle missing values, and construct sequential time windows for the acquired PhysioNet dataset.
Baseline Training	Code standard *FedAvg* on PyTorch locally to establish stable forecasting baselines.
FPDAF Integration	Embed *CUSUM* threshold triggering and *Attention* layers into the personalized model heads.
Federated Simulation	Deploy 3-5 independent simulated Hospital "Clients" to test concept drift adaptation latency.
Evaluation Metrics	Benchmark system performance utilizing **Accuracy, Precision, Recall, F1-Score, and AUC-ROC**.

Execution Plan (Components & Flow)

Subdomain	Matching Components Identified	Execution Target
Data Engineering	Custom ICU Sequence Windows, Imputation matrices.	Preprocessing
Deep Learning Base	Extending LSTM with a **novel** Temporal Attention constraint.	Global Model
Decentralization	**Extended** *FedAvg* integrated with Attention-Aware weighting.	Federated Loop
Customization	*FedPer* baseline modified dynamically via **FPDAF** heads.	Client Local Model
Statistical Math	*CUSUM* mathematically altered to act as a Neural Network freezing layer trigger.	Drift Tracker

Novel Solution Pipeline: Extract Data $\rightarrow$ Modify *FedAvg* Aggregation $\rightarrow$ Inject Attention Constraints $\rightarrow$ Sync Modified Shared Layers $\rightarrow$ Trigger Novel *CUSUM*-Adaptive Local Reset.

FPDAF Detailed Implementation Logic (What & How)

- **1. Decentralization (Global Syncing)**
  - *What:* Establish a privacy-preserving Baseline.
  - *How:* Code a central PyTorch Server script to securely aggregate only the `Global Temporal Base` gradient weights across 3-5 simulated hospital nodes, ensuring raw patient telemetry never leaves the client device.
- **2. Customization (Personalized Modeling)**
  - *What:* Adapt Sepsis predictions to specific hospital demographics.
  - *How:* Instantiate independent `Local Personalization Heads` (*FedPer*) completely disconnected from the *FedAvg* loop, allowing PyTorch client layers to heavily fit their local ICU distributions independently.
- **3. Statistical Math (Active Concept Drift)**
  - *What:* Enable active AI response to rapid patient deterioration.
  - *How:* We will sequentially track prediction residuals (e_t). If the $S_t > H$ CUSUM mathematical threshold breaches due to sudden physiological crashing, we dynamically unfreeze the personalization head for instant local patient recalibration!

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Thank You Questions?